

Gender Differences in Job Application Strategies: An Experimental Investigation*

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Abstract

Job-seekers may react strategically to their beliefs about a job opening's "competitiveness": the quantity and quality of people applying to it. We experimentally investigate how gender differences in such beliefs may make women less likely to apply for high-paying yet competitive jobs. We design a job application game where members of a group are either ranked randomly or based on their performance on a math and science test. Without knowing their rank, they select one of three "jobs": two yield large payoffs to the best-ranked player who selects them, while the third guarantees a small payoff to all who select it. We find that willingness to apply to a job decreases in the perceived competitiveness of its applicant pool, but is more sensitive to the believed quality of the pool than the believed size thereof. Furthermore, gender differences in these beliefs create gaps in willingness to apply to the highest-paying competitive job. Our findings highlight when and how policies that reveal the competitiveness of a job's applicant pool (e.g., by disclosing the current number of applications) may increase the proportion of qualified female applicants.

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1 Introduction

Despite significant progress in women’s educational attainment and career advancement, gender earnings gaps continue to emerge early in workers’ careers and grow as their careers progress (Blau and Kahn, 2017; Bertrand et al., 2010). Recent work has investigated how gender differences in job search behaviour may contribute to the early onset and persistence of these gaps. This work has treated job search as an individual decision-making problem, showing that men assign less value to non-pecuniary job characteristics (Burbano et al., 2024; Fluchtmann et al., 2024; Collis and Van Effenterre, 2024; Le Barbanchon et al., 2021; Wiswall and Zafar, 2018) and are more optimistic about prospective job offers (Cortés et al., 2023).

In this paper, we argue that job search happens in a strategic setting: an individual’s likelihood of receiving a job offer depends on the degree to which they need to compete against other applicants. Willingness to apply to a job may therefore depend on the perceived *competitiveness* of its applicant pool: the perceived quantity and quality of other applicants. A large experimental literature suggests that men and women’s perceptions of competitiveness may differ. Women often assess themselves more pessimistically than men of equal ability (Exley and Nielsen, 2024; Exley and Kessler, 2022; Bordalo et al., 2019; Coffman, 2014), and may thus perceive themselves as lower-quality applicants. Additionally, review articles highlight that women tend to be more risk-averse (Eckel and Grossman, 2008) and competition-averse (Niederle and Vesterlund, 2011). As a result, women may respond more strongly to their beliefs about the quantity of other applicants, or hold different beliefs altogether.

We explore job search behaviour in a controlled experimental setting that isolates the impact of strategic considerations on willingness to apply to a job. Our experiment is designed to address two main questions. First, how do individuals’ beliefs about the quantity and quality of a job’s applicant pool impact their willingness to apply to it? Second, to what extent do gender differences in these beliefs create gender gaps in willingness to apply to high-paying, competitive jobs? Our core experimental task is a novel job application game inspired by field evidence on job search patterns. We begin by summarizing this evidence, which comes from a survey of over 2,000 upper-year students and alumni from a large public university. A majority of respondents reveal that, when applying to a job, their beliefs about the quantity and quality of the job’s applicants are primary considerations. Furthermore, nearly three-quarters of respondents list the time cost of the application as a primary consideration. We therefore incorporate all of these factors into the design of our job application game.

Participants in our experiment complete two versions of this game, which differ in the criterion used to rank members of a five-person group. In the “base game”, members are ranked randomly, while in the “ability game”, they are ranked based on their performance on a math and science test. Without knowing their rank, each member must select one of three “jobs”: lotteries that either yield a large payoff to the best-ranked group member who selects them (Jobs A and B), or guarantee a small payoff (Job C). We refer to Jobs A and B as the competitive jobs. Limiting the choice to one job creates an opportunity cost to applying, which captures the empirically-perceived importance of an application’s time cost. The difference in the competitive jobs’ payoffs is proportional to the early-career gender earnings gap, with Job A paying more to its highest-ranked applicant than Job B. A participant’s

willingness to apply to a higher-paying job may therefore depend on their beliefs about the quantity and quality of other participants applying to it. To determine the impact of these beliefs on participants' choices, we collect quantitative data on the beliefs themselves, as well as qualitative data on whether those beliefs influenced participants' job choices.

We begin our analysis by establishing the effects of participants' beliefs about a job's competitiveness on their willingness to apply to it. On the extensive margin, we find that up to 80% of participants' job choices are influenced by these beliefs. That said, job choice is more sensitive to beliefs about the relative quality of a job's applicants than to beliefs about the quantity of people applying. More precisely, the former beliefs influence nearly twice as many participants' job choices. On the intensive margin, we find that participants' willingness to apply to a job decreases in the perceived competitiveness of its applicant pool. Specifically, a participant is more likely to apply to a given competitive job if they believe the other has a larger applicant pool, and is more likely to apply to a higher-paying job if they believe they have performed better on the math and science test. This first set of results thus highlights the importance of strategic considerations in job search behaviour.

Having established the effect of beliefs about jobs' competitiveness on job choices, we next investigate the presence and impact of gender differences in these beliefs. In the base game, we find suggestive gender differences in beliefs about jobs' competitiveness, with women being more likely to believe that Job A has more applicants. In the ability game, we find no gender differences in confidence, which we define as the difference between believed and true test performance. Moreover, gender differences in beliefs create noteworthy gender differences in job choices. In the base game, where we *do* observe gender differences in beliefs, men are 52% more likely to apply to Job A, and 45% less likely to apply to Job B. In contrast, there are no gender differences in job choice in the ability game, where we *do not* observe gender differences in beliefs.

Taken altogether, our results highlight the importance of strategic considerations in job search behaviour. These findings offer insight on the conditions under which revealing the competitiveness of a job's applicant pool may encourage women to apply to it. Prior work has shown that disclosing the size of a job's applicant pool (Gee, 2019) and reducing ambiguity surrounding its required qualifications (Coffman et al., 2024) increase the share of women who apply to it. Our findings suggest that such policies are effective because they help correct women's pessimistic beliefs about the likelihood of obtaining a competitive position. This in turn implies that such policies would be most effective in male-dominated or -stereotyped domains, where men and women may be most likely to hold different beliefs about jobs' competitiveness.

This research contributes to two main strands of literature. First, we provide a deeper understanding of the consequences of gender differences in beliefs and preferences on economic outcomes. Prior work has linked gender differences in risk and competition preferences to gender differences in test performance (Baldiga, 2014), educational and career choices (Buser et al., 2014; Flory et al., 2015) and financial decision making (Barber and Odean, 2001). Our work is most related to an emerging literature that connects gender differences in behavioural attributes to job search behavior. While it is known that men and women tend to accept different job offers (Cortés et al., 2023), we show that

gender gaps may emerge even earlier in the job search process: men and women may apply to different jobs to begin with.

Second, we add to a body of work on the behavioural determinants of job search patterns. Previous research has shown that unemployed workers' job search patterns are affected by reference-dependent thinking (DellaVigna et al., 2017, 2022), risk preferences (Pannenberg, 2010), time preferences (DellaVigna and Paserman, 2005) and biased beliefs about employment prospects (Spinnewijn, 2015). We show that strategic considerations are yet another important factor for many applicants, who respond rationally to their beliefs the competitiveness of an applicant pool.

The rest of this paper is structured as follows. Section 2 outlines the empirical evidence that motivates the design of the job application game. This game was the core task in the experiment, whose design is described in Section 3. Section 4 presents the experimental results. Section 5 concludes by discussing the interpretation and implications of these results.

2 Empirical Evidence

The design of our job application game is motivated by a survey of 2,048 upper-year students and alumni from a large public university, which investigated factors that influence individuals' job preferences (Collis and Van Effenterre, 2024). The researchers kindly allowed us access to the data from a subset of their survey questions, wherein respondents rated the influence of the following factors on their decision to apply to a job:

1. Their beliefs about the *quantity* of other applicants. Respondents were asked, "When applying to a job, with what frequency do you consider the number of other people that may be applying to that job?"
2. Their beliefs about the *quality* of other applicants. Respondents were asked, "When applying to a job, with what frequency do you consider how your qualifications may compare to those of others who may be applying to that job?"
3. The time cost associated with the application. Respondents were asked, "When applying to a job, with what frequency do you consider the time associated with completing an application process? (for instance: writing a new cover letter, updating your CV, etc.)"

Responses were recorded using a seven-point scale, whose options included "Never", "Rarely (less than 10% of the time)", "Occasionally (about 30% of the time)", "Sometimes (about 50% of the time)", "Frequently (about 70% of the time)", "Usually (about 90% of the time)" and "Every time". We define a factor as being a primary consideration in a respondent's job application decisions if they consider it at least 50% of the time.

About half of respondents (50.8%) indicate beliefs about the size of the applicant pool to be a primary consideration when applying to jobs. A larger share of respondents (77.8%) reveal beliefs about their relative qualifications to be a primary consideration. While this field data shows that beliefs about

jobs’ competitiveness are frequently considered, it is unclear how and to what extent these beliefs influence an individual’s decision to apply to a job. An incentivized laboratory experiment, in which beliefs and job choices can be linked, can help identify this intensive-margin effect. Lastly, most respondents (72.2%) list an application’s time cost as a primary consideration. This is consistent with the view that such costs are non-negligible, and suggests that time-constrained undergraduates may only have the capacity to apply to a subset of possible job openings, rather than every job they might be interested in. Faced with this opportunity cost, they may prioritize their applications to specific job openings, perhaps prioritizing jobs for which they believe they have a strong chance of being hired.

3 Experimental Design

The core task in our experiment is a job application game that embodies the empirical patterns outlined in Section 2. We also elicit individuals’ beliefs about jobs’ competitiveness, to investigate whether individuals respond strategically to these beliefs. In this section, we discuss the main components and structure of the experiment. The complete experimental instructions are presented in Appendix A.

3.1 The Job Application Game

To separately identify the effects of beliefs about quantity versus quality of jobs’ applicant pools on job application decisions, participants play two versions of a job application game. In each version, participants are randomly assigned to groups of five and are assigned a rank within their group. In the version we call the “base game”, rank is determined randomly. Each group member is randomly assigned a number from one to five, with each member being equally likely to be assigned each number. In the version we call the “ability game”, the ranking is based on members’ relative performance on a math and science test, which is described in greater detail in Section 3.2.

In both versions of the game, members of a group simultaneously choose one of three lotteries, without knowing their rank. We refer to the lotteries as “jobs” to make the decision less abstract and more similar to the job application decisions that motivate our study. Limiting the choice to one job captures the application time costs discussed in Section 2. Two of the jobs are competitive, yielding a large payoff to the best-ranked group member who selects them. The ratio of the competitive jobs’ payoffs to that member is similar to the 10-13% gap in men and women’s early-career earnings ([Cortés et al., 2023](#)). The third job is non-competitive, guaranteeing a small payoff to all who select it. More precisely, the job options include:

- Job A, which pays the best-ranked applicant \$20 and all other applicants \$1.
- Job B, which pays the best-ranked applicant \$17 and all other applicants \$1.
- Job C, which pays all applicants \$6.

In the base game, a rational player’s optimal strategy depends on their risk preferences and their beliefs about the number of group-mates who will apply to each competitive job. More risk-averse play-

ers should select the non-competitive job. All other players' likelihood of selecting a given competitive job should decrease in the number of their group-mates they believe will apply to it. In the ability game, a rational player's optimal strategy additionally depends on their beliefs about their relative performance on the math and science test. A player's likelihood of selecting a higher-paying competitive job should increase in their believed relative test performance, given they are increasingly likely to be the best-ranked player who selects that job.

3.2 Structure of the Experiment

The experiment consists of five stages: a math and science test, two job application games, belief elicitation tasks, a risk preference elicitation task, and a survey. The order in which participants completed each stage is shown in Figure 1. Before stages one through three, participants were required to pass a comprehension quiz to ensure they understood the instructions for the task they would complete in that stage.

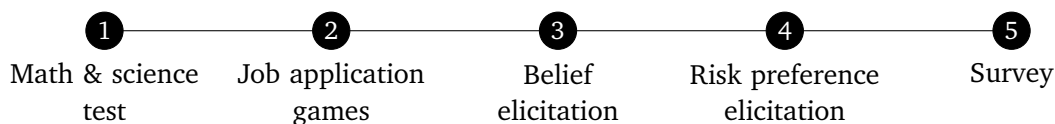


Figure 1: Structure of the Experiment

In the first stage, participants completed a math and science test. The test consisted of 20 multiple choice questions selected from the General Science, Arithmetic Reasoning and Mathematics Knowledge sections of the Armed Services Vocational Aptitude Battery's (ASVAB) bank of questions. The ASVAB is a test traditionally used by the United States Armed Forces to assess the knowledge and skills of potential enlistees. We chose these sections of the ASVAB as the source of our test questions because they have been found to generate similar test scores across genders in previous experiments (Exley and Kessler, 2022; Exley and Nielsen, 2024), but also because its math and science foundations are primarily stereotyped as male-dominated subject areas (Coffman, 2014). Participants had thirty seconds to answer each question, and the order of questions was varied between participants. The median participant answered 15 of the 20 questions correctly.

After all members of a group completed the test, their ranks in each job application game were determined. If multiple members of a group received the same test score, their ranks were determined according to test completion time, with a faster time resulting in a better rank. The second stage consisted of the job application games, whose order was randomized between participants.

In the third stage, we elicited participants' beliefs about the competitiveness of jobs' applicant pools.¹ First, we elicited their beliefs about their own quality, relative to their group-mates'. Participants indicated, in percentage terms, how likely they thought they were to have achieved each rank (first through

¹We could have elicited participants' beliefs about their group-mates' strategies either before or after the job application games. That said, the act of eliciting beliefs before a game tends to impact the strategy participants end up playing (Erev et al., 1993; Croson, 1999, 2000; Rutström and Wilcox, 2009; Gächter and Renner, 2010). Given that we did not want to prime participants to play any given strategy, we conducted the belief elicitation tasks after the job application games.

fifth) on the math and science test. We incentivized these beliefs using the binarized scoring rule (Hos-sain and Okui, 2013), keeping the payment rule’s description clear and simple (Danz et al., 2022). To investigate the source of any distortions in perceived quality, we also elicited participants’ beliefs about their test score. Finally, we elicited participants’ beliefs about the quantity (i.e., the number) of their group-mates who applied to each competitive job in each job application game. To incentivize the latter two sets of beliefs, we used a simple scheme that paid \$4 if a participant’s guess was correct and \$0 otherwise.

In the fourth stage, we elicited participants’ risk preferences using the Gneezy and Potters (1997) investment task. Finally, participants completed a survey wherein they are reminded of the job they chose in each job application game, and must describe the rationale behind their choice in each game. Participants answered these questions in open text, with no constraints on subject matter or length of response.

3.3 Implementation

The experiment was programmed and deployed in oTree (Chen et al., 2016), and conducted in-person² at the Toronto Experimental Economics Laboratory (TEEL). Participants were recruited from the University of Toronto student body³ using ORSEE (Greiner, 2015). We conducted eight sessions, each with either 15 or 20 participants, for a total of 155 participants. At the beginning of each session, participants completed a demographic questionnaire that included a question regarding their gender identity. Participants were then randomly assigned to a group of five, subject to the constraint of ensuring the greatest possible gender balance in each group.

Participants’ total payment from the experiment included a \$7 show-up fee, a variable payment from one task from stage 1 through 3, and a bonus payment from the risk preference elicitation task. The task that counted for a participant’s variable payment was randomly determined at the beginning of the experiment and revealed to them upon completion of the experiment. The median participant earned \$17.57 and took 27 minutes to complete the experiment.

4 Results

We begin our analysis by establishing the impact of participants’ beliefs about jobs’ competitiveness on their job choice. Having established this effect, we then determine whether these beliefs differ by gender, and finally, whether we observe gender differences in job choices. Our final sample consists of

²In the context of our experimental design, there are several advantages to conducting the experiment in-person. First, it mitigates the risk of cheating on the math and science test. Believing that members of their group are cheating could impact participant beliefs about their relative ranking: for instance, it could make them less likely to believe they have achieved a high rank. Second, being able to physically observe their potential group-mates makes the competitive nature of their decisions more salient. Finally, given that participant payments depend on their group-mates’ decisions, conducting in-person sessions allows all members of a group to participate simultaneously, and thus get paid immediately at the end of the session. Participants tend to take experimental tasks more seriously if they know they will be paid immediately (Charness et al., 2023).

³Given that we are studying a potential cause of the early-career gender earnings gap, conducting the experiment with members of the target population increases external validity.

153 people, 45% of whom identify as women. An additional two people completed the experiment, but were excluded as they did not identify as either a man or a woman. All reported p -values are from two-sided tests.

4.1 Impact of beliefs on choices

We seek to identify the impact of beliefs on choices along both the extensive and intensive margins. That is, which participants' job choices are influenced by their beliefs about jobs' competitiveness (extensive margin), and how do those beliefs influence these participants' job choices (intensive margin)? The results from [Collis and Van Effenterre \(2024\)](#) suggest that beliefs about a job's competitiveness will impact some, but not all, participants' willingness to apply to a job. We use the survey responses to identify those participants. Two undergraduate students from the University of Toronto were hired as research assistants to independently categorize the responses to these survey questions. Their primary tasks⁴ were twofold: (1) identify whether, in the base game, a participant's job choice was influenced by their beliefs about their group-mates' job choices (2) identify whether, in the ability game, a participant's job choice was influenced by their beliefs about their performance on the test. The research assistants had not participated in the experiment, and were not privy to our research questions. There is a high degree of consensus between the research assistants' independent classifications: out of 153 participants, they disagree on only nine participants' responses to each survey question. Furthermore, their categorizations are highly positively correlated ($\rho = 0.88$ for the base game, $\rho = 0.84$ for the ability game, $p < 0.01$ for both). In all analyses in this section, we classify a participant as having considered the quantity or quality of a job's applicants if at least one research assistant classifies them as having done so. However, our results are qualitatively similar when using either research assistant's classification.

Table 1: Share of Individuals Considering Jobs' Competitiveness (Extensive Margin Effect)

	Proportion		p -value of difference
	Lab (1)	Field (2)	
Considers applicant quantity	0.425	0.508	0.05
Considers applicant quality	0.797	0.778	0.58

Notes. Columns 1 and 2 present the shares of our experimental participants (in Column 1) and [Collis and Van Effenterre \(2024\)](#)'s survey respondents (in Column 2) who consider applicant quantity or quality when deciding whether to apply to a job. Column 3 presents p -values from proportions tests.

Table 1 reports the share of individuals whose beliefs about jobs' competitiveness influence their willingness to apply to a job. We report the results from our lab experiment (in column 1) and, for comparison purposes, the job preference survey from [Collis and Van Effenterre \(2024\)](#) (in column 2). In our experiment, just under half of participants consider the quantity of other applicants applying to

⁴See Appendix B for the complete instructions for the categorization tasks.

a job, while just under 80% consider how their qualifications compare to others'. These shares are very similar to those from [Collis and Van Effenterre \(2024\)](#). The share of individuals considering applicant quality does not differ significantly across settings, and a slightly larger share of survey respondents are influenced by their beliefs about the quantity of other applicants (proportions test, $p = 0.05$). This small difference may have resulted, for instance, from the different classification schemes used in the two settings. Thus, consistent with empirical trends, a large share of participants' job choices are influenced by their beliefs about jobs' competitiveness. That said, but again consistent with empirical evidence, beliefs about the quality of other applicants are nearly twice as impactful as beliefs about the quantity thereof.

Table 2: Impact of Beliefs on Job Choices (Intensive Margin Effect)

	Applies to Job A		Applies to Job B		Applies to Job C	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Base game						
Believes other competitive job has (weakly) ⁵ more applicants	0.201 (0.1370)		0.174 (0.1456)		- -	- -
Believed # of applicants at other competitive job		0.110 (0.073)		0.181** (0.084)	- -	- -
Constant	0.511*** (0.168)	0.476** (0.192)	0.214* (0.127)	-0.016 (0.168)	- -	- -
Observations	65	65	65	65	-	-
Panel B: Ability game						
Believed rank	-0.326*** (0.034)		0.054 (0.038)		0.272*** (0.033)	
Believed score		0.074*** (0.012)		-0.005 (0.013)		-0.069*** (0.012)
Constant	1.446*** (0.108)	-0.445** (0.210)	-0.0683 (0.118)	0.131 (0.206)	-0.377*** (0.106)	1.313*** (0.213)
Observations	122	122	122	122	122	122

Notes. The table cells present the estimated coefficients from OLS regressions of a column's dependent variable on a row's independent variable. All regressions include controls for the session and the order in which a participant played the job application games. Robust standard errors, clustered at the participant-level, are in parentheses. * indicates $p < 0.10$, ** indicates $p < 0.05$ and *** indicates $p < 0.01$.

Having established how many participants are influenced by their beliefs about jobs' competitiveness, we next determine the impact of those beliefs on these participants' job choices. As we are interested in the intensive margin effect, we restrict the sample to participants whose beliefs about jobs' competitiveness influence their job choice. Table 2 reports how the likelihood of applying to a given job changes depending on the beliefs a participant holds.

⁵If one believes that Job B has weakly more applicants than Job A and is risk-seeking enough to apply to one of the competitive jobs, applying to Job A is a strictly dominant strategy. However, a rational player who applies to Job B must believe that Job A has more applicants. Thus, in the regressions for which "Applies to Job A" is the dependent variable, this dependent variable indicates the belief that Job B has weakly more applicants. However, in the regressions for which "Applies to Job B" is the dependent variable, this dependent variable indicates the belief that Job A has strictly more applicants.

In the base game, we investigate the impact of participants' beliefs about applicant quantity on their job choice. The results are summarized in Panel A of Table 2. As shown in columns 1 and 3, believing one job is relatively more competitive makes a participant less likely to apply to it. Participants who believe Job B has at least as many applicants as Job A are 20 percentage points more likely to apply to Job A. Similarly, participants who believe Job A has more applicants than Job B are 17 percentage points more likely to apply to Job B. The results from columns 2 and 4 indicate that this pattern holds when competitiveness is defined in absolute terms. The likelihood a participant applies to Job B increases by 18 percentage points for each additional believed applicant to Job A. Similarly, for each additional believed applicant to Job B, the likelihood a participant applies to Job A increases by 11 percentage points. Although we are under-powered to determine the statistical significance of these effects, the large point estimates suggest there is, at minimum, a negative relationship between the believed competitiveness of a job and the likelihood a participant applies to that job. Phrased slightly differently, believing a given job has a larger applicant pool - whether measured in absolute or relative terms - makes a participant more likely to apply to a different job.

In the ability game, we investigate the impact of participants' beliefs about their own quality on their job choice. The results are summarized in Panel B of Table 2. We assess the impact of believed own-quality both in relative terms (believed rank in one's group) and in absolute terms (believed score out of 20). We define a participant's believed rank as the median of the distribution of their believed probability of having achieved each rank. Note that an increase in believed rank indicates a decrease in believed relative performance. As a result, column 1 indicates that, if believed relative performance increases by one rank, a participant is almost 33 percentage points more likely to apply to the highest-paying job (Job A). For that same one-rank increase, column 5 indicates participants are 27 percentage points less likely to apply to the lowest-paying, non-competitive job (Job C). Columns 2 and 6 show a similar pattern when quality is defined in absolute terms. For each additional believed correct answer on the test, participants are 7 percentage points more likely to apply to Job A, and 7 percentage points less likely to apply to Job C. Therefore, a participant is more likely to apply to higher-paying jobs, and less likely to apply to lower-paying jobs, if they believe themselves to be a higher-quality applicant. Taken altogether, the results from Panels A and B suggest that, conditional on being influenced by their beliefs about jobs' competitiveness, participants respond rationally to those beliefs when deciding which job to apply to.

4.2 Gender differences in beliefs

In Section 4.1, we established that many participants are influenced by their beliefs about jobs' competitiveness when applying to jobs, and that those who are react strategically to those beliefs. As a result, if we observe gender differences in beliefs about jobs' competitiveness in one of the job application games, we would expect men and women to choose different jobs in that game. In this Section, we thus determine whether men and women hold different beliefs about jobs' competitiveness in either of the job application games. As we are interested in gender differences in job choices driven by gender differences in beliefs, for the purpose of this analysis, we restrict the sample to participants who strategically

respond to their beliefs about jobs' competitiveness.

Table 3: Beliefs about Quantity of Other Applicants and Own Quality, by Gender

	<u>Proportion or mean</u>		p -value of difference
	Men	Women	
	(1)	(2)	(3)
Panel A: Base game			
Believes Job A has more applicants	0.279 (0.068)	0.364 (0.103)	0.49
Believed # of Job A applicants	1.628 (0.115)	1.818 (0.204)	0.33
Believed # of Job B applicants	1.698 (0.131)	1.591 (0.142)	0.46
Panel B: Ability game			
Rank belief error	0.260 (0.161)	0.235 (0.164)	0.90
Score belief error	-1.014 (0.327)	-0.939 (0.402)	0.72

Notes. Columns 1 and 2 display the proportion (in the first row) or mean (in all other rows) of the variable in the corresponding row, with standard errors in parentheses. Column 3 displays the 2-sided p -value from a two-sample test of proportions (in the first row) or Wilcoxon rank-sum test (in all other rows). Samples in each panel consist of the participants whose beliefs about jobs' competitiveness (in the base game) or their qualifications (in the ability game) influence their job choice (22 women and 43 men in Panel A; 49 women and 73 men in Panel B). A participant's rank belief is the median of the distribution of their belief about the rank they achieved within their group. Belief errors are defined such that a positive error indicates overconfidence.

Table 3 displays measures of participants' beliefs about jobs' competitiveness for each job application game, separated by gender. Panel A, which is specific to the base game, summarizes participants' beliefs about the size of jobs' applicant pools. Given that roughly half of our participants are influenced by these beliefs, we are slightly underpowered for the base game results. Nonetheless, we observe suggestive gender differences in participants' beliefs about the quantity of their group-mates who applied to each competitive job. 36% of female participants believe Job A has a larger applicant pool, while only 28% of male participants hold the same belief. Furthermore, relative to the average female participant, the average male participant believes there to be an additional 0.1 applicants to Job A and a fewer 0.2 applicants to Job B. While these results are statistically insignificant (proportions tests, $p \in [0.33, 0.49]$), they are economically meaningful. All results indicate that women are more likely than men to believe that Job A is most competitive, and the gender difference in beliefs is as large as 30.5%.

Panel B, which is specific to the ability game, summarizes participants' beliefs about their own quality. Given that we control for true performance when investigating job choices in Section 4.3, we also control for it in this analysis of participants' beliefs. More precisely, we investigate gender differences in belief errors. These errors provide a measure of confidence, and are defined as the difference between a participant's true and believed performance. Both men and women tend to be overconfident in their

rank and underconfident in their test scores, and the magnitude of their belief errors is not significantly different (proportions tests, $p \in [0.72, 0.90]$). Furthermore, the small (at most 9.6%) gender differences in each type of belief error point in opposite directions, with men being slightly more overconfident in their rank yet more underconfident in score. These results suggest that there are no meaningful differences in confidence across gender groups.⁶

4.3 Gender differences in job choices

Thus far, we have shown that a meaningful share of participants react strategically to their beliefs about jobs' competitiveness. We have also identified one setting in which these participants' beliefs differ by gender, and another in which they do not. The final step of our analysis thus establishes whether the behaviour of this strategic sub-sample may create a gender job choice gap in the overall sample. If so, we would expect more men to apply to the higher-paying, competitive job when they are more likely to believe it is less competitive (i.e., in the base game), but no gender differences in job choice when men and women hold similar beliefs about jobs' competitiveness (i.e., in the ability game).

⁶The lack of a gender confidence gap in our experiment is somewhat surprising, given the prevalence of such gaps in male-stereotyped tasks (Bordalo et al., 2019; Coffman, 2014). One potential explanation for this difference relates to demographic differences between participant pools. Most participants in earlier studies attended high school in the United States (see Table 1 in both Bordalo et al. (2019) and Coffman (2014)). While we did not ask our participants how long they had been living in North America, a large fraction of the University of Toronto's student population - and thus TEEL's participant pool - is from outside of North America. Given that cultural background affects gender stereotypes (see, e.g., Gneezy et al. (2009)), it is possible that many of our participants did not perceive the math and science test as a task where men would have a performance advantage.

Table 4: Gender Differences in Job Choices

	Applies to Job A		Applies to Job B	
	(1)	(2)	(3)	(4)
Panel A: Base game				
Male	0.1819** (0.0825)	0.05397 (0.1094)	-0.1711** (0.0774)	-0.0785 (0.0912)
Considers applicant quantity		-0.1965* (0.1179)		0.4044*** (0.1297)
Male $\times$ considers applicant quantity		0.3357** (0.1635)		-0.3335** (0.1596)
Constant	0.3486*** (0.1104)	0.4226*** (0.1188)	0.3792*** (0.1207)	0.2210* (0.1299)
Panel B: Ability game				
Male	0.0971 (0.0901)	-0.1420 (0.2043)	0.0411 (0.0753)	0.0691 (0.1188)
Considers applicant quality		-0.1541 (0.1323)		0.2300** (0.1044)
Male $\times$ considers applicant quality		0.2998 (0.2172)		-0.0594 (0.1547)
Constant	0.0490 (0.2454)	0.1685 (0.2583)	-0.0409 (0.1777)	-0.1379 (0.1990)
Observations	153	153	153	153

Notes. The table cells present the estimated coefficients from OLS regressions of a column's dependent variable on a row's independent variable. All regressions include controls for the session and the order in which a participant played the job application games. The regressions in Panel B additionally include a control for a participant's test score. Robust standard errors, clustered at the participant-level, are in parentheses. * indicates $p < 0.10$, ** indicates $p < 0.05$ and *** indicates $p < 0.01$.

Table 4 reports how the likelihood of applying to a given job changes depending on a participant's gender. We focus first on the base game, whose results are presented in Panel A. A basic regression of job choice on gender reveals that, in the base game, men are significantly more likely to apply to the higher-paying competitive job. The coefficient estimates in columns 1 and 3 indicate that men are 18 percentage points more likely than women to apply to Job A, and 17 percentage points less likely than women to apply to Job B. Crucially, however, this effect is entirely driven by strategic participants. In columns 2 and 4, we report the results of regressions that interact gender with an indicator for considering the size of an applicant pool when deciding which job to apply to. In these regressions, the coefficient on gender becomes statistically insignificant, while the interaction coefficients are significant and much larger than the gender coefficients from the basic regression. Phrased differently, men and women who do not consider the number of other applicants to a job are equally likely to apply to Jobs A and B. It is only within the group of participants who consider the quantity of jobs' applicants - who display suggestive gender differences in beliefs about these quantities - that gender differences in job choices emerge.

Next we turn to the ability game, whose results are presented in Panel B. Across all regression specifications, we find no gender differences in the likelihood of applying to either competitive job. These results are consistent with the fact that, in the ability game, there are no gender differences in strategic participants' beliefs about their own (and by extension, other applicants') quality. Taken altogether, the results from both job application games suggest that gender differences in beliefs about jobs' competitiveness can create gaps in willingness to apply to high-paying, competitive jobs, even if only a subset of applicants are influenced by these beliefs.

5 Conclusion

Using a lab experiment, we show that many individuals approach job search decisions strategically, and identify the consequences of gender differences in job search strategies. We find that willingness to apply to a job decreases in the perceived competitiveness of its applicant pool: believing a job has a larger number of applicants, or that those applicants are more highly-qualified, makes many individuals less likely to apply to it. That said, application decisions are more sensitive to beliefs about the relative quality of a job's applicants than to beliefs about the quantity of people applying. Finally, we show that gender differences in these beliefs create gaps in willingness to apply to high-paying, competitive jobs. In a setting where women are more likely to believe that the highest-paying job is most competitive, they are less likely to apply for it. However, men and women are equally likely to apply to that job in a setting where there are no gender differences in beliefs about jobs' competitiveness.

While we frame our main experimental task as a job application game, our results also shed light on application decisions that occur later in workers' careers. A worker's likelihood of being promoted often depends on their willingness to apply for more prestigious, yet potentially more competitive, career advancement opportunities. For instance, an academic researcher's beliefs about the caliber or number of other papers submitted to a highly-ranked journal may impact her willingness to submit her own paper to it. Our results thus point to a channel through which gender earnings gaps may not only emerge, but also magnify as careers progress.

Our job application game aims to capture the core features of job application decisions. That said, it may be valuable to study how other common features of the job-search process may influence job application strategies. First, players in our job application games know their group size, and thus the maximum number of applicants to each job. In reality, however, job-seekers are often uncertain about how large a job's applicant pool may become. Second, while our participants know relatively little about their group-mates, job-seekers may have a signal of the quantity and quality of a job's applicants. For instance, an undergraduate student may know that most students in her program tend to apply for positions at a particular company, but that she is ranked within the top 10% of students in her graduating class. Third, given that job search often involves multiple rounds of applications and offers, job-seekers may adapt their strategies in response to the outcome of previous application attempts. Studying the impact of such informational frictions, as well as how search strategies evolve in a dynamic setting, may be promising avenues for future research.

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Appendix

A Experimental Instructions

Welcome to the experiment!

General instructions. This is an experiment designed to study decision-making. If you pay close attention to the instructions that follow, you can earn a significant amount of money. Please ensure all of your electronics (cell phones, smart watches, etc.) are put away. Also, please do not talk with others during the experiment.

Structure of the experiment. The experiment consists of four parts. Across the first three parts, there are five tasks. The fourth part includes an additional short task and a survey. You will receive detailed instructions prior to each part and task.

Payment. Your payment consists of three components:

- *Task that counts.* One of the five tasks from the first three parts of the experiment has been randomly selected to count for your payment. The task that counts will only be revealed to you at the end of the experiment. This payment protocol suggests that **you should treat each task in the first three parts as if it is the task that counts.**
- *Additional task.* You may receive additional earnings based on your choice in the additional short task in the fourth part of the experiment.
- *Show-up fee.* You will receive \$7.00 for completing the experiment.

Questionnaire. We want to know more about you! Prior to beginning the experiment, please complete this brief questionnaire.

- What degree are you currently pursuing? [Options: Bachelor's degree, Master's degree, Doctoral degree, Other degree (not listed)]
- In what year did you begin the degree you are currently pursuing? [Options: 2023, 2022, 2021, 2020, 2019, 2018, 2017 or earlier]
- What is your primary field of study (e.g., Anthropology, Chemistry, Mechanical Engineering, etc.)? [Text field where participant types answer]
- How old are you? [Text field where participant types answer]
- What is your gender? [Options: Female, Male, Non-binary, Other (not listed), Prefer not to answer]

Part 1 - Instructions

Part 1 consists of one task, which is a 20-question test that will assess your aptitude in general science, arithmetic reasoning and math. You will be presented with each of the 20 questions on a separate page,

and will have up to 30 seconds to answer each question. You may click "Next" to submit your answer to a question before the 30 seconds are over.

Payment: If this is the task that counts, you will earn \$1.00 per test question you answer correctly. Before proceeding to the test, you will be asked a comprehension question to ensure you have understood these instructions.

Part 1 - Comprehension Quiz

If you answer the question correctly, you will be taken to the next page. If you answer the question incorrectly, an error message will pop up and ask you to try again.

Question 1

If this task is randomly selected as the task that counts, your additional payment

- (A) will not depend on how many questions you answer correctly on this test.
- (B) will be lower if you answer more questions correctly on the test.
- (C) will depend on your answer to one randomly-selected question from the test.
- (D) will be higher if you answer more questions correctly on the test.

Part 1 - Begin Test

Click the "Next" button to begin the test. Good luck!

AUTHORS' NOTE

Participants completed 20 multiple-choice questions taken from the General Science, Arithmetic Reasoning and Mathematics Knowledge sub-tests of the Armed Services Vocational Aptitude Battery (ASVAB). Questions were presented to each participant in a random order. Below we show one of the 20 questions.

Question n of 20

Remaining time (seconds): [remaining time shown]

Air is less dense than water because

- ☐ it is lighter.
- ☐ its molecules are farther apart.
- ☐ its molecules are closer together.
- ☐ it moves more quickly and easily.

End of Part 1

Congratulations! You have now completed Part 1 of the experiment. Please wait while the other subjects complete the test. Afterwards, we will proceed to Part 2.

Part 2 - General Instructions

Part 2 consists of two tasks with the same general structure. You have been randomly grouped with four other subjects who are sitting in the lab right now, and will complete each task as a member of this five-person group. In each task, each group member (yourself included) will be assigned a number from 1 through 5, called their "rank", with 1 being the best possible rank and 5 being the worst possible rank. **The rank assignment protocol depends on the task**, and you will be informed of the relevant protocol prior to each task.

In each of the Part 2 tasks, you must choose one of three payment options. The potential payment and degree of competition for that payment varies across the three options. We will describe this choice as "applying" to one of the three "Jobs" described below, which each have different "hiring" policies. However, being hired at a Job simply involves receiving the payment associated with that Job; it does not involve performing any additional tasks. All applicants that are not hired receive \$1.00.

- *Job A* hires only the best-ranked group member who applies to it, and pays them \$20.00.
- *Job B* hires only the best-ranked group member who applies to it, and pays them \$17.00.
- *Job C* hires all applicants, and pays them each \$6.00.

Neither you, nor any of your group-mates, will know their rank when choosing which Job to apply to. Your decision may therefore depend on what you think your rank is, as well as the choices you think your group-mates will make.

Payment: If one of the Part 2 tasks is the task that counts, you will be paid according to the choices you and your group-mates made in that task.

Examples: Suppose you apply to Job A and your rank is 3.

- Example 1. Suppose your group-mates with ranks 4 and 5 also apply to Job A, and your group-mates with ranks 1 and 2 apply to Job B. You would be the best-ranked applicant to Job A in this case. Therefore, you would be hired at Job A, and would receive the \$20.00 payment associated with being hired at this Job.
- Example 2. Suppose all of your group-mates also apply to Job A. You would not be the best-ranked applicant to Job A in this case. Therefore, you would not be hired at Job A, and would receive the \$1.00 payment associated with not being hired.

Before proceeding to the tasks, you will be asked some comprehension questions to ensure you have

understood these instructions.

Part 2 - Comprehension Quiz

If you answer the question correctly, you will be taken to the next page. If you answer the question incorrectly, an error message will pop up and ask you to try again.

Question 1

Suppose you apply to Job A and your rank is 2. If your group-mate with rank 1 also applies to Job A,

- (A) you will be hired at Job A.
- (B) you will not be hired at Job A.

Part 2 - Comprehension Quiz

If you answer the question correctly, you will be taken to the next page. If you answer the question incorrectly, an error message will pop up and ask you to try again.

Question 2

Suppose you apply to Job B and your rank is 5. You will be hired at Job B

- (A) regardless of whether any of your group-mates apply to Job B.
- (B) only if none of your group-mates apply to Job B.

Part 2 - Comprehension Quiz

If you answer the question correctly, you will be taken to the next page. If you answer the question incorrectly, an error message will pop up and ask you to try again.

Question 3

Suppose you apply to Job C and your rank is 3. If your group-mates with ranks 1 and 2 also apply to Job C,

- (A) you will be hired at Job C, as Job C hires all who apply.
- (B) you will not be hired at Job C, as Job C hires only the best-ranked applicant.

Part 2 - Comprehension Quiz

If you answer the question correctly, you will be taken to the next page. If you answer the question incorrectly, an error message will pop up and ask you to try again.

Question 4

If you are hired at a Job, you will receive the payment associated with that Job

- (A) without having to perform any additional tasks.

(B) only if you successfully perform some additional task.

AUTHORS' NOTE

The order in which the job application games were presented to a participant was randomly determined at the beginning of the experiment. Below we show one of the two possible orders.

Part 2 - Task 1 Instructions

In this task, ranks are assigned **randomly**. That is, we assigned ranks by randomly assigning a number from 1 through 5 to each member of your group (yourself included). Each member was equally likely to be assigned each rank, and no two members were assigned the same rank.

Before proceeding to the task, you will be asked a comprehension question to ensure you have understood this rank assignment protocol.

Part 2, Task 1 - Comprehension Quiz

If you answer the question correctly, you will be taken to the next page. If you answer the question incorrectly, an error message will pop up and ask you to try again.

Question 1

In the Part 2 tasks, each member of your group is assigned a rank from 1 to 5. In Task 1, these ranks are assigned

- (A) via group deliberation.
- (B) by a participant in another group.
- (C) randomly, with each member being equally likely to be assigned each rank.

Part 2 - Task 1

Your task	Your choice
<i>Please choose one of the following actions. Ranks were assigned randomly. Click here for a summary of the general Part 2 task instructions.</i> • Apply to Job A. The best-ranked applicant to Job A receives \$20.00; all other applicants receive \$1.00. • Apply to Job B. The best-ranked applicant to Job B receives \$17.00; all other applicants receive \$1.00. • Apply to Job C. All applicants to Job C receive \$6.00.	Please make your choice: [Options: Apply to Job A, Apply to Job B, Apply to Job C]

Part 2 - Task 2 Instructions

In this task, ranks are assigned **according to performance on the test from Part 1**. That is,

- the member of your group with the highest score has been assigned rank 1,
- the member of your group with the second-highest score has been assigned rank 2,
- ...
- the member of your group with the lowest score has been assigned rank 5.

If two or more group members received the same score, these members' test completion times will be used to break the tie (with a faster completion time resulting in a better rank). Due to this tie-breaking rule, you do not need to worry about multiple group members having the same rank.

Before proceeding to the task, you will be asked a comprehension question to ensure you have understood this rank assignment protocol.

Part 2, Task 2 - Comprehension Quiz

If you answer the question correctly, you will be taken to the next page. If you answer the question incorrectly, an error message will pop up and ask you to try again.

Question 1

In the Part 2 tasks, each member of your group is assigned a rank from 1 to 5. In Task 2, these ranks are assigned based on group members'

- (A) order of arrival at the experiment (1 = first, 5 = last).
- (B) relative scores on the test from Part 1 (1 = highest score, 5 = lowest score).
- (C) birth order (1 = oldest, 5 = youngest).

Part 2 - Task 1

Your task	Your choice
<i>Please choose one of the following actions. Ranks were assigned based on Part 1 test scores (with a higher test score resulting in a better rank). Click here for a summary of the general Part 2 task instructions.</i> • Apply to Job A. The best-ranked applicant to Job A receives \$20.00; all other applicants receive \$1.00. • Apply to Job B. The best-ranked applicant to Job B receives \$17.00; all other applicants receive \$1.00. • Apply to Job C. All applicants to Job C receive \$6.00.	Please make your choice: [Options: Apply to Job A, Apply to Job B, Apply to Job C]

End of Part 2

Congratulations! You have now completed Part 2 of the experiment. Please wait while the other subjects make their decisions. Afterwards, we will proceed to Part 3.

Part 3 - General Instructions

Part 3 consists of two tasks that involve guessing quantities related to Parts 1 and 2. You will receive detailed instructions and payment information prior to each task.

Part 3 - Task 1 Instructions

In this task, you will guess how your performance on the test from Part 1 ranked in your group of five. Specifically, you will guess how likely it is that you were ranked first, second, third, fourth or fifth in your group. You will do this by entering a percentage between 0 and 100 for each possible rank, with the five percentages adding up to 100. Recall that a higher relative score resulted in a better rank, and no members received the same rank due to the tie-breaking rule.

Payment: If this is the task that counts, we will reward the accuracy of your guess with a payment rule that secures the **highest chance of winning \$15.00 when you provide your best guess**. To maximize your chance of winning, you should thus assign a 0% likelihood to any rank you don't think you achieved, and a positive likelihood to any rank you think you may have achieved, with a higher likelihood for a rank you believe to be more likely. You may view an example of how the payment rule works by clicking this button: [Payment Rule Example](#)

Examples: You will enter your guess in a table that looks like those in the examples below.

Example 1		Example 2	
Possible rank	Percent chance	Possible rank	Percent chance
You ranked 1st	0%	You ranked 1st	0%
You ranked 2nd	100%	You ranked 2nd	0%
You ranked 3rd	0%	You ranked 3rd	0%
You ranked 4th	0%	You ranked 4th	75%
You ranked 5th	0%	You ranked 5th	25%
Total	100%	Total	100%

The guess shown in Example 1 puts a 100% likelihood on the second-place rank, and a 0% likelihood on every other rank. This guess reflects a confident belief that one has ranked second. The guess shown in Example 2 puts a 75% likelihood on the fourth-place rank, a 25% likelihood on the fifth-place rank, and a 0% likelihood on every other rank. This guess reflects a belief that it is quite likely that one has ranked fourth, but it is also possible - although much less likely - that one has ranked fifth. Both are potential valid guesses, as in each example, the five percentages add up to 100.

Before proceeding to the task, you will be asked a comprehension question to ensure you have understood these instructions.

Part 3, Task 1 - Comprehension Quiz

If you answer the question correctly, you will be taken to the next page. If you answer the question incorrectly, an error message will pop up and ask you to try again.

Question 1

In Task 1 of Part 3, you will guess how your performance on the test from Part 1 ranked in your group of five. Submitting your best guess will maximize your chance of winning \$15.00.

- (A) True
- (B) False

Part 3 - Task 1

Think back to your **performance on the test in Part 1**. In the table below, please indicate **how likely you believe it is that you achieved each rank, relative to your group-mates**. You have the highest chance of winning \$15.00 when you provide your best guess.

Possible rank	Percent chance
You ranked 1st	☐ %
You ranked 2nd	☐ %
You ranked 3rd	☐ %
You ranked 4th	☐ %
You ranked 5th	☐ %
<i>Total</i>	☐ %

Part 3 - Task 2 Instructions

In this task, you will guess the following quantities:

- The number of questions you answered correctly on the test from Part 1.
- For the Part 2 task where ranks were randomly assigned,
 - The number of your group-mates that applied to Job A.
 - The number of your group-mates that applied to Job B.
- For the Part 2 task where ranks were assigned based on relative test scores,
 - The number of your group-mates that applied to Job A.
 - The number of your group-mates that applied to Job B.

You will do so by entering a number that represents your best guess of each of these five quantities.

Note that you are asked to guess how many of your group-mates applied to Jobs A and B in each Part 2 task. That is, if you applied to Job A or B in one of the Part 2 tasks, you must guess how many other group members also applied to that Job in that task.

Payment: If this is the task that counts, we will reward the accuracy of your guess by paying you \$4.00 per quantity you guess correctly.

Example: Suppose you correctly guess the number of questions you answered correctly on the test, as well as the number of your group-mates that applied to Job A in each Part 2 task. However, suppose you incorrectly guess the number of your group-mates that applied to Job B in each Part 2 task. Given that you made three correct guesses and two incorrect guesses, your earnings would be $\$4.00 \times 3 = \12.00 . Before proceeding to the task, you will be asked a comprehension question to ensure you have understood these instructions.

Part 3, Task 2 - Comprehension Quiz

If you answer the question correctly, you will be taken to the next page. If you answer the question incorrectly, an error message will pop up and ask you to try again.

Question 1

In Task 2 of Part 3, you will guess the number of your group-mates who applied to Jobs A and B in each Part 2 task. If you applied to one of these Jobs in a given Part 2 task, the number you guess should

- (A) include you.
- (B) exclude you.

Part 3 - Task 2

Please enter your best guess of each of the five quantities below. You will receive \$4.00 per quantity you guess correctly.

Think back to the **test from Part 1**.

- How many questions (out of 20) do you think you answered correctly? [Options: list of numbers from 0 to 20]

Think back to the **Part 2 tasks**. Click [here](#) for a reminder of Job A and B's hiring policies, as well as the choice you made in each task.

In the task where **ranks were randomly assigned**,

- How many of your group-mates do you think applied to Job A? [Options: list of numbers from 0 to 4]
- How many of your group-mates do you think applied to Job B? [Options: list of numbers from 0 to 4]

In the task where **ranks were assigned based on relative test scores**,

- How many of your group-mates do you think applied to Job A? [Options: list of numbers from 0 to 4]
- How many of your group-mates do you think applied to Job B? [Options: list of numbers from 0 to 4]

End of Part 3

Congratulations! You have now completed Part 3 of the experiment. Please wait while the other subjects complete the tasks. Afterwards, we will proceed to Part 4.

Part 4 - General Instructions

Part 4 consists of a short additional task and a survey. Prior to each, you will receive detailed instructions and, if applicable, payment information.

Part 4 - Additional Task

In this task, you must decide how much of a \$2.00 budget to invest in a lottery. The remaining amount (i.e., money not invested in the lottery) is yours to keep. The lottery gives you a $\frac{2}{3}$ (67%) chance to lose your investment, and a $\frac{1}{3}$ (33%) chance to win two-and-a-half times your investment. Your investment must be rounded to the nearest 5 cents. Your earnings from this task will be added to your total earnings from the experiment.

I will invest [Options: list of dollar amounts from \$0 to \$2, increasing in \$0.05 increments]

Survey

Think back to the **tasks from Part 2**.

1) In the task where your rank was assigned **randomly**, you chose to apply to Job A. Please describe why you made that choice. [Text field where participant types answer]

2) In the task where your rank was assigned **based on your relative test performance**, you chose to apply to Job A. Please describe why you made that choice. [Text field where participant types answer]

Results

Thank you for participating in this experiment. Here are your earnings from the task that counts and the additional task:

- *Task that counts*: [description of the task that counted and the participant's earnings from it]
- *Additional task*: [description of participant's earnings from the risk preference elicitation task]

Your total payment from the experiment is therefore \$X (\$7.00 show-up fee + \$Y from the task that counts + \$Z from the additional task). An experimenter will come by your desk shortly and pay you this amount in cash.

Thanks for participating!

B Instructions for Survey Categorization

AUTHORS' NOTE

The research assistants received separate sets of instructions for the Base and Ability Game survey categorization tasks. Sections 1 and 2.1 were common to each set of instructions. Sections 2.2 and 3 differed; the content of the sections specific to each task are under separate sub-headings.

1 Background: Our Experiment

We ran an experiment where subjects played two versions of a job application game in groups of five. In one version (the “Base Game”), group members were ranked randomly. In the other (the “Ability Game”), they were ranked based on their performance on a math and science test. Without knowing their rank, all members of a group had to simultaneously choose one of these three payment options, which we called “jobs”:

- *Job A.* The best-ranked applicant to Job A received \$20; all other applicants received \$1.
- *Job B.* The best-ranked applicant to Job B received \$17; all other applicants received \$1.
- *Job C.* All applicants to Job C received \$6.

In Section 2.2, we sometimes refer to Jobs A and B as the “competitive” jobs. After completing these application games, subjects answered two survey questions:

- **Base Game Survey Question:** “In the task where your rank was assigned randomly, you chose to apply to Job [A/B/C]. Please describe why you made that choice.”
- **Ability Game Survey Question:** “In the task where your rank was assigned based on your relative test performance, you chose to apply to Job [A/B/C]. Please describe why you made that choice.”

2 Your Task: Survey Categorization

2.1 Overview

We will give you a spreadsheet whose rows each refer to one participant. The participant’s ID, Ability Game job choice and Ability Game survey answer are indicated in the first three columns of the spreadsheet. The headers of the subsequent columns refer to categories of survey answers.

Your task is to identify whether a participant’s answer to the **Ability Game Survey Question** belongs to certain categories. You will do so by going through a list of categorization questions for each participant (starting with Question 1), and typing the number 1 in the columns corresponding to categories that the answer belongs to. Some questions may not be relevant to a participant’s answer; Section 2.2 includes instructions on which questions you may skip, depending on the participant’s response.

A participant may refer to their answer to the other survey question. For instance, they may say, “My reasoning is the same as for the other game.” To help you in such cases, we will give you a second

spreadsheet that includes the participant's answer to the other survey question. That spreadsheet will also include subjects' IDs, so you can find the participant's answer to the other survey question by searching their ID.

If you have any questions while you are completing this task, you can email us at alexandra.ballyk@mail.utoronto.ca and annabel.thornton@mail.utoronto.ca.

2.2 Questions for Categorization

BASE GAME

QUESTION 1. Did the participant consider their group-mates' job choices when making their decision?

- If yes, put a 1 in column `s_base1`.
- If no, leave column `s_base1` blank.

Here are some examples of things a participant might say if they considered their group-mates' job choices:

1. "I thought most people would take the risk of applying for Job A, so I applied to Job B."
2. "If others chose Jobs A or B, then there's a chance I would only get \$1 if I chose one of those jobs, so I chose Job C."

If you put a 1 in column `s_base1`, proceed to Question 2a. Otherwise, proceed to Question 3.

QUESTION 2A. Which of the competitive jobs did a participant believe would have the fewest applicants?

- If they indicated that they believe Job A will have the fewest applicants, put a 1 in column `s_base2a_Alesscomp`.
- If they indicated that they believe Job B will have the fewest applicants, put a 1 in column `s_base2a_Blesscomp`.
- If they did not indicate which of Jobs A or B they believe will have the fewest applicants, leave the `s_base2a` columns blank.

Here are some examples of things a participant might say if they believe one of the competitive jobs has fewer applicants:

1. "I thought most of my group-mates would apply to Job A."
2. "I thought my group-mates would play it safe and apply to Jobs B or C."

The categories associated with Question 2a are mutually exclusive: you may put a 1 in at most one of the `s_base2a` columns. After you have addressed Question 2a, proceed to Question 2b.

QUESTION 2B. Why did a participant believe their group-mates would apply to specific jobs?

- If they discussed their group-mates' risk preferences or expressed the belief that their group-mates would choose a safe or risky job, put a 1 in column `s_base2b_riskprefs`.
 - Example: "I think my risk-loving group-mates applied to Job A, and my risk-averse group-mates applied to Job C."
- If they expressed the belief that their group-mates would select a job because it may pay a certain amount, put a 1 in column `s_base2b_pmtsize`.
 - Example: "I thought fewer people would select Job B because it pays less than Job A."
- If they gave a reason other than the reasons above, put a 1 in column `s_base2b_other`.
 - Example: "I thought people chose Job A because they like the letter A."
- If they did not indicate why they believed their group-mates would apply to specific jobs, leave the `s_base2b` columns blank.

If a participant lists more than one of the above reasons, put a 1 in the `s_base2b` column that you think corresponds to the *main* reason. After you have addressed Question 2b, proceed to Question 4.

QUESTION 3. What is the participant's rationale for their job choice?

- If they referred to their own risk preferences, the safeness/riskiness of a certain choice, or their probability of having achieved a certain rank, put a 1 in column `s_base3_riskprefs`.
 - Examples: "Job C was the safest option", "I had an equal chance of being assigned each rank, so I took the risk and applied for Job A"
- If they referred to the amount of money a certain job may pay, put a 1 in column `s_base3_pmtsize`.
 - Example: "Might as well go for the job with the highest potential reward"
- If they gave a reason other than the above two reasons, type the number 1 in the column labelled `s_base3_other`.
 - Example: "I randomly decided between Jobs A and B"

If a participant lists more than one of the above rationales, put a 1 in the `s_base3` column that you think corresponds to their *primary* rationale. After you have addressed Question 3, proceed to Question 4.

QUESTION 4. Does the participant regret their choice?

- If yes, put a 1 in column `s_base4`.
- If no, leave column `s_base4` blank.

Here are some examples of things a participant might say if they regret their choice:

1. "I applied to Job B because I thought most people would want the highest-paying job. But now I regret my choice because I realized others might've been thinking the same thing."
2. "I chose Job B by mistake; I meant to apply to Job A."

After you have addressed Question 4, proceed to the next participant.

ABILITY GAME

QUESTION 1. Did the participant consider their performance on the test when making their decision?

- If yes, put a 1 in column `s_ability1`.
- If no, leave column `s_ability1` blank.

Here are some examples of things a participant might say if they considered their performance on the test:

1. "I applied to Job A because I think I did very well on the test."
2. "I'm not sure how my test performance compared to my group-mates', so I applied to Job C to be safe."

After addressing Question 1, proceed to Question 2.

QUESTION 2. Did the participant consider their group-mates' job choices when making their decision?

- If yes, put a 1 in column `s_ability2`.
- If no, leave column `s_ability2` blank.

Here are some examples of things a participant might say if they considered their group-mates' job choices:

1. "I thought the people who did well on the test would apply to Job A, so I applied to Job B."
2. "I think others applied to Jobs B or C because they weren't confident in their scores, making Job A the best choice for me."

If you put a 1 in column `s_ability2`, proceed to Question 3a. If you did not put a 1 in column `s_ability2` but you put a 1 in column `s_ability1` when you addressed Question 1, proceed to Question 5. If you left columns `s_ability2` and `s_ability1` blank, proceed to Question 4.

QUESTION 3A. Which of the competitive jobs did the participant believe would have the fewest applicants?

- If they indicated that they believe Job A will have the fewest applicants, put a 1 in column `s_ability3a_Alesscomp`.
- If they indicated that they believe Job B will have the fewest applicants, put a 1 in column `s_ability3a_Blesscomp`.
- If they did not indicate which of Jobs A or B they believe will have the fewest applicants, leave the `s_ability3a` columns blank.

Here are some examples of things a participant might say if they believe one of the competitive jobs has fewer applicants:

1. "I predicted that most people would choose Jobs A or C."
2. "Most people probably weren't confident enough to choose Job A."

The categories associated with Question 3a are mutually exclusive: you may put a 1 in at most one of the `s_ability3a` columns. After you have addressed Question 3a, proceed to Question 3b.

QUESTION 3B. Why did the participant believe their group-mates would apply to specific jobs?

- If they discussed their group-mates' beliefs about their scores, or mentioned that their group-mates might be over- or under-confident in their scores, put a 1 in column `s_ability3b_confidence`.
 - Example: "I think many of my group-mates believed they did the best on the test, which lead them to apply to Job A."
- If they discussed their group-mates' risk preferences or expressed the belief that their group-mates would choose a safe or risky job, put a 1 in column `s_ability3b_riskprefs`.
 - Example: "Many people probably played it safe and opted for Job C's sure payment."
- If they expressed the belief that their group-mates would select a job because it may pay a certain amount, put a 1 in column `s_ability3b_pmtsize`.
 - Example: "I think many of my group-mates chose Job A because it pays the most."
- If they gave a reason other than the reasons above, put a 1 in column `s_ability3b_other`.
 - Example: "I think many of my group-mates chose Job C because they wanted to let others have a chance at earning the most money."
- If they did not indicate why they believed their group-mates would apply to specific jobs, leave the `s_ability3b` columns blank.

If a participant lists more than one of the above reasons, put a 1 in the `s_ability3b` column that you think corresponds to the *main* reason. After you have addressed Question 3b, proceed to Question 5.

QUESTION 4. What is the participant's rationale for their job choice?

- If they referred to their own risk preferences or the safeness/riskiness of a certain choice, put a 1 in column `s_ability4_riskprefs`.
 - Examples: “Job C was the safest option”, “I took the risk and applied to Job A”
- If they referred to the amount of money a certain job may pay, put a 1 in column `s_ability4_pmtsize`.
 - Example: “I chose Job A because I wanted the \$20”
- If they gave a reason other than the reasons above, type the number 1 in the column labelled `s_ability4_other`.
 - Example: “I randomly decided between Jobs A and B”

If a participant lists more than one of the above rationales, put a 1 in the `s_ability4` column that you think corresponds to their *primary* rationale. After you have addressed Question 4, proceed to Question 5.

QUESTION 5. Does the participant regret their choice?

- If yes, put a 1 in column `s_ability5`.
- If no, leave column `s_ability5` blank.

Here are some examples of things a participant might say if they regret their choice:

1. “I applied to Job B because I thought most people would want the highest-paying job. But now I regret my choice because I realized others might’ve been thinking the same thing.”
2. “I chose Job B by mistake; I meant to apply to Job A.”

After you have addressed Question 5, proceed to the next participant.

3 Pre-Meeting Task

Before our next meeting, we would like you to think about how you would categorize two sample responses. For each response below, please go through the questions outlined in Section 2.2 (starting with Question 1) to determine which columns of the spreadsheet you would put a 1 in. We will discuss your categorization during the meeting.

BASE GAME

- **Sample response #1:** “I figured most of my group-mates would choose the job that offers the highest possible payment (Job A), which made Job B the best choice for me.”
- **Sample response #2:** “After thinking about my chances of being assigned each rank, I concluded that Job C was the safest option.”

ABILITY GAME

- **Sample response #1:** "I wasn't confident that I had the best test score, but I figured my group-mates felt the same way, which meant they probably applied to Jobs B and C. That made Job A the best choice for me."
- **Sample response #2:** "I thought people were going to play it safe and apply to Job C, so I applied to Job B. But now I think I should've applied to Job A instead."